INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION
BATTERIES, RECHARGEABLE, NICKEL CADMIUM SEALED

This specification is approved for use by U.S. Army Research Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable, sealed rechargeable nickel cadmium storage batteries. Detailed electrical and physical requirements for individual batteries are specified on specification sheets.

1.2 Classification.

1.2.1 Type designation. The type designation of nickel cadmium, sealed rechargeable batteries shall be in the following form (see 3.1).

BB-	586	/U
Component	Battery number	Installation indicator
(1.2.1.1)	(1.2.1.2)	(1.2.1.3)

1.2.1.1 Component. Rechargeable batteries are identified by the two-letter symbol "BB" followed by a hyphen.

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, New Jersey 07703-5601, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1.2 Battery type number. The battery type number identifies the basic design of the battery (see 3.1) and consists of a three digit number.

1.2.1.3 Installation indicator. The installation indicator identifies equipment the battery is used in, i.e. /TAS or if "universal", i.e. /U indicates use in various equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

Federal

FF-S-92	Screws, Machine, Slotted, Crossed Recessed or Hexagon Head.
QQ-S-571	Solder, Tin Alloy, Lead-Tin Alloy & Lead Alloy.
QQ-S-763	Steel Bars, Wire, Shapes, and Forgings, Corrosion-Resisting

Military

MIL-M-14	Molding Plastics and Molded Plastic Parts Thermosetting.
MIL-M-13231(ER)	Marking of Electronic Items.
MIL-F-14256	Flux, Soldering, Liquid, Paste Flux, Solder Paste and Solder Paste Flux (For Electronic/Electrical Use) General Specification For
MIL-W-16878	Wire, Electrical, Insulated, General Specification for.
MIL-P-19904	Plastic Sheets, Acrylonitrile-Butadiene Styrene Terpolymer, Rigid.
MIL-M-43719	Marking Materials and Markers, Adhesive, Elastomeric, Pigmented, General Specification For.

STANDARDS

Federal

FED-STD-595	Colors.
FED-STD-601	Rubber, Sampling and Testing.

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Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals.
MIL-STD-970	Standards and Specifications, Order Of Preference For The Selection Of
MIL-STD-45662	Calibration System Requirements.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 484	Steel Products, Wrought Stainless and Heat Resisting, General Requirements for.
ASTM D 2000	Rubber Products in Automotive Applications Classification System for.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications or specification sheets), the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

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3.1.1 Selection of specification and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.2 First Article requirements. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.8) in accordance with 4.4.

3.3 Parts, materials, and processes. Parts, materials, and processes shall be such as to enable the battery to meet its performance and other requirements, as specified herein. Rating and derating values shall conform to Requirement 18 of MIL-STD-454. When deemed necessary by the government, certification from the source of the part, material, or component shall be required (see 4.3.1 and 6.2). When a definite material or component is specified, it shall be in accordance with the applicable specification, drawing or requirement listed in Table I.

3.3.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be employed in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889 and shall meet Requirement 16 of MIL-STD-454.

3.4 Visual and mechanical. Batteries shall be inspected to meet the requirements of 3.4.1 through 3.4.9, 3.5 and 3.14 (see 4.5.2.2).

3.4.1 Hardware. Hardware shall be in accordance with Table I and the following subparagraphs.

3.4.1.1 Riveting. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. The distance from the centers of rivet holes to the edges of the material in which the holes are formed shall be at least 1-1/2 times the rivet diameter. After riveting, the joints shall be tight, the jointed parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.4.1.2 Mounting screws for connector. Unless otherwise specified (see 3.1) two (2) mounting screws shall be furnished with the battery connector. They shall be corrosion resistant steel per ASTM A 484. Mounting screws shall be tightened so as not to impede the sideways movement of the connector.

3.4.2 Pressure release device. Unless specified otherwise in the applicable specification sheets a pressure release device shall be incorporated into the battery. This device shall be designed to release below the bursting pressure of the container and above the normal operating pressure of the battery. The device is intended to function only as an emergency release to protect the battery against abnormal pressure build-up. The design of the battery shall be such that the paths from the individual cell units to the battery vent are sufficient to permit the removal of gases (see 4.11).

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TABLE I. Materials and Components.

Materials or components	Requirements paragraph	Inspection Method
Case (Battery)	3.4.3, (see 3.1)	
Solder	QQ-S-571, MIL-STD-454, 3.4.6, 3.4.7.2	
Soldering flux	MIL-F-14256, 3.4.6	
Connector	3.4.5, (see 3.1)	
Metals	3.3.1, 3.8.1, MIL-STD-454	
Pressure Release Device (Battery)	3.4.2, (see 3.1)	
Pressure Release Device (Cell)	3.5.3	
Machine Screws	FF-S-92, QQ-S-763, (see 3.1)	
Molding plastics and molded plastic parts	MIL-M-14	
Wire	3.4.6, MIL-W-16878	
Elastomers	3.8.2, ASTM D 2000	4.7, FED-STD-601
Cells	3.5, 3.5.1 thru 3.5.4	

3.4.3 Battery case. Battery case shall be capable of maintaining the specified dimensions during the life of the battery. The surfaces of cases shall have a smooth finish free from pitting, blow-holes, rough spots, or other deformations. The case shall be fabricated of material having sufficient strength to withstand deformation. Case shall be acrylonitrile butadiene, styrene (ABS), or equal.

3.4.4 Marking. Each battery shall be provided with a permanent, legible hot-stamped or printed identification label. Markings shall be in accordance with the requirements of MIL-M-13231(ER). Direct printing on the battery is the preferred method.

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3.4.4.1 Identification label. The identification label shall be as shown on the applicable specification sheets (see 3.1).

3.4.4.2 Date of manufacture code. The code shown shall indicate the month and year of manufacture of the battery by means of a four-digit number in which the first two digits shall indicate the number of the month and the last two digits shall indicate the year. Months earlier than the tenth month shall be a single digit preceded by "O". When a battery is completed during the last three working days of a month or the first three working days of the subsequent month, the manufacturer is permitted to use either month as the coded month of manufacture.

3.4.4.3 Terminal marking. Positive terminals shall be legibly indicated by +, and negative terminals may be identified by -. The polarity markings shall be placed as close as possible to the applicable side of the receptacle. A schematic drawing shall be placed as close as possible to the receptacle as shown on the applicable specification sheet.

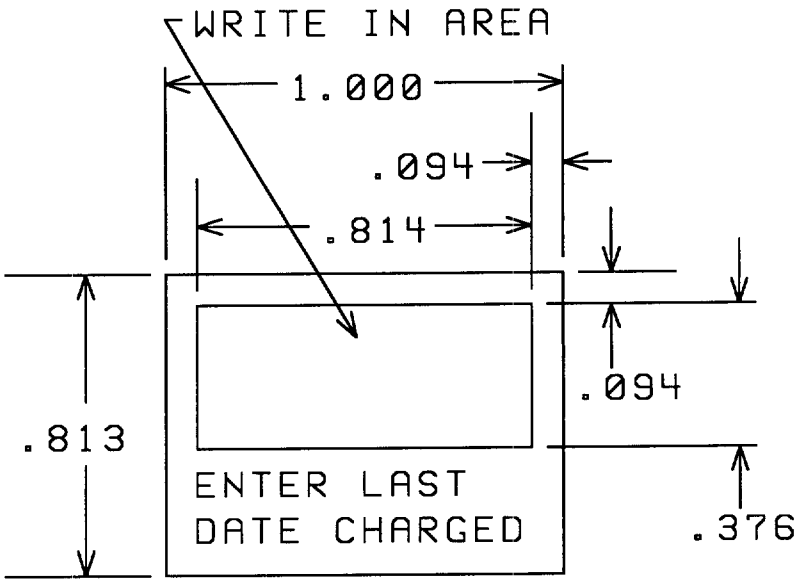
3.4.4.4 Writable charge history label. A writable label shall be furnished per Figure 1 in the location shown on the applicable specification sheet (see 3.1).

3.4.4.5 Instructions. Each battery shall be furnished with complete instructions for operation and charging of the battery either on a label, printed on the battery or furnished with the battery per applicable specification sheet. Direct printing on the battery is the preferred method.

3.4.5 Connector. Battery connector shall be as shown on the applicable specification sheet.

3.4.6 Wire and soldering. Soldering shall conform to Requirement 5 of MIL-STD-454 except that the liquid flux shall be type R or RMA of MIL-F-14256 and flux-cored solder shall be type R or RMA of QQ-S-571. Wire shall be per MIL-W-16878.

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NOTES:

1. ALL DIMENSIONS IN INCHES.

2. MATERIAL SHALL BE "WRITABLE" VINYL .005-.010 THICK, WITH A PERMANENT PRESSURE SENSITIVE ADHESIVE BACK PER MIL-M-43719/1,2, 3, TYPE 2. "WRITE-IN" AREA SHALL BE WHITE ON BLACK BACKGROUND. THIS SURFACE MUST BE CAPABLE OF BEING WRITTEN ON IN PENCIL AND ERASED REPEATEDLY, WITHOUT DAMAGE.

3. LETTERING SHALL BE WHITE AND ALL CAPITALS.

4. TOLERANCES: DECIMALS $\pm$.030.

FIGURE 1. Writable Label

3.4.7 Cleaning.

3.4.7.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely.

3.4.7.2 Battery. After fabrication, batteries shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material.

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3.4.8 Color. The color of the battery container shall be lusterless green, closest to chip No. 34087 per FED-STD-595 or as otherwise stated on individual battery drawings or on the applicable specification sheet (see 3.1).

3.4.9 Battery condition for shipping. The battery shall be furnished in a discharged state (see 6.3.2) with instructions attached to each battery and a dust cap or cover over the receptacle.

3.5 Cells. The battery shall be made up of the required number of individual series connected cells which have demonstrated their capability to meet the specified electrical requirements.

3.5.1 Cell balance. The capacities of the individual cells used in each battery shall not vary by more than 5% from the average capacity value of the cells in that particular battery (see 4.3.2).

3.5.2 Protective coverings. Batteries shall be fabricated of cells with individual metal containers which have a protective coating or jacket of a minimum 0.003" thickness.

3.5.3 Cell pressure release device. A resealable pressure release device shall be incorporated into each cell. This device shall be designed to release well below the bursting pressure of the cell container and appreciably above the nominal operating pressure. The device is intended to function only as an emergency release to protect the cells against abnormal pressure build-up.

3.5.4 Cell leakage. Individual cells used in the batteries shall be free of electrolyte leakage indicated by the presence of white carbonate deposits at the vent or closure areas of the cells. The seals between cell terminals and cell covers shall be so constructed that the batteries assembled from these cells shall be capable of meeting the electrical leakage requirements and the cycle life requirements on the applicable specification sheet.

3.6 Dimensions and weights. The dimensions and weights of batteries shall be as shown on the applicable specification sheets. Dimensions shall be gauged or measured accurately to determine conformance (see 4.6.6).

3.7 Attitude. The battery shall be designed for operation in any position and shall meet the specified capacity, voltage and electrolyte leakage requirements (see 3.1).

3.8 Chemical requirements.

3.8.1 Corrosion resistance. External parts of the battery shall show no evidence of cracking, pitting, chipping, scaling, corrosion or other deleterious effects after being tested as specified herein.

3.8.2 Resistance of elastomer materials. All elastomer materials used in the battery shall meet ASTM D 2000 requirements and show no cracks, blisters or other deterioration, nor cause degradation of battery performance during any of the tests specified herein.

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3.9 Temperature requirements.

3.9.1 Operating. The battery shall meet the temperature performance requirements of the applicable specification sheet (see 3.1).

3.9.2 Non-operating. The battery shall show no evidence of breaking or cracking of the case or cover when subjected to thermal shock temperature cycling in the range of 160°F (71°C) to -75°F (-59°C) (see 4.9.1).

3.10 Electrical requirements.

3.10.1 Electrical leakage. Unless otherwise specified (see 3.1), no potential in excess of 0.5 volts shall be obtained from the case or non-connected terminal to any terminal when tested in accordance with 4.10.

3.10.2 Battery voltage. When charged briefly (4.5.2.3), the battery voltage at 10 seconds of charge shall be no greater than the number of cells per battery, (N), times 1.5 V/cell nor less than N times 1.2 V/cell (see 4.5.2.3).

3.11 Performance. The discharge current in amperes for each discharge shall be equal to the nominal rated capacity (C-rating) in ampere-hours multiplied by the rate factor shown in the applicable specification sheet. The temperature of discharge and the minimum end voltage shall be the value listed in the applicable specification sheet. Inability to meet the specified minimum requirements shall constitute a failure. Should the battery meet the test requirements, the specified test schedule shall be resumed.

3.12 Environmental test requirements. The battery when subjected to the tests listed in Table II that are required by the applicable specification sheet shall show no:

a. Dimensional distortion beyond specified limits.

b. Sharp current overvoltage fluctuations during any charge or discharge period.

c. Mechanical failure of any part.

d. Failure to mate with a corresponding part.

e. Diminution of the rated capacity specified in the applicable specification sheet following the performance of each test.

f. Electrical leakage (3.10.1).

g. Safety hazard.

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TABLE II. Environmental tests.

Tests	Test Paragraph
Mechanical Shock	4.9.2
Vibration	4.9.3
Thermal Shock	4.9.1
Overcharge	4.8.1.1

3.13 Immersion. The batteries shall show no evidence of leakage of water into cases or leakage of gas from the cases after being subjected to the immersion or alternate leakage tests of 4.9.4.

3.14 Workmanship. Batteries shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect their life, serviceability, or appearance in accordance with Requirement 9 of MIL-STD-454.

3.15 Battery case expansion. The battery shall be designed to prevent dimensional expansion that will prevent its removal from equipment during and after battery case venting. The battery shall not require more than 12 pounds of force for removal when tested per Battery Case-Vent Test (see 4.11).

3.16 Battery storage life. The battery shall be capable of at least 5 years of warehouse storage during which storage temperatures may vary between the limits of -10°C (14°F) to 40°C (104°F). The battery when removed from storage shall show no:

a. Dimensional distortions beyond the specified limits.

b. Mechanical failure of any part.

c. Evidence of marking, pitting, chipping, scaling, corrosion or other deleterious effects on external parts of battery.

d. Electrical leakage (3.10.1).

e. Cell leakage (3.5.4).

f. Inability to provide the rated capacity specified in the applicable specification sheet after a reconditioning charge at C/10 for 16 hours followed by discharge per 4.8.2.1.

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3.17 Battery safety. If required to insure safety under conditions of overcharge and/or misuse, the battery, unless otherwise specified, shall contain at a minimum a normally closed thermoswitch that shall open at 70° ± 5°C and close at 60 ± 5°C. Protective devices employed shall not adversely affect battery performance or life. The details of any safety protection devices shall be disclosed to and approved by the Government.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a. Inspections covered by subsidiary documents including components and materials inspection (see 4.3 and 4.3.1).

b. First article inspection (see 4.4).

c. Quality conformance inspection (see 4.5).

4.3 Inspections covered by subsidiary documents. Inspections required under the applicable subsidiary documents form a part of the inspections required by this specification.

4.3.1 Samples of components and materials. For those items that are not referenced by a subsidiary specification, three samples of components and materials, treated and processed as they would be in the finished battery shall be inspected.

4.3.2 Cell inspections. The contractor shall establish and maintain a system for measuring and

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recording the capacities of individual cells (or subassemblies of cells) showing that they meet the requirements of 3.5.1.

4.4 First article inspection. This inspection shall be performed by the contractor unless otherwise specified in the contract. It shall consist of inspection and testing specified in Table III and shall be performed in the order indicated. If a different test is indicated on the applicable specification sheet (see 3.1) perform that test in place of the established test. Production may start after the samples successfully complete the 112th cycle of the life cycle test. For batteries with cycle requirements less than 112 cycles, all the cycles must be completed prior to start of production.

4.4.1 First article compliance report. The contractor shall submit a compliance report with any battery type that is submitted for acceptance. The report, complete in detail, shall be submitted regardless of whether the tests to determine compliance with specification requirements are conducted at the contractor's facilities, government facilities or independent facilities. The contractor's report shall indicate to what degree the equipment complies with each individual electrical, mechanical, or other specification requirement. The degree of compliance shall be expressed in the report by the use of the same terms as those used in the specification to express the requirement (see 6.8).

4.4.2 Sampling plan. Unless specified, the contractor shall furnish six first article samples selected at random of each battery type on order.

4.4.2.1 Cycling schedule. Two samples (nos. 5 and 6) shall be given the cycle life test of 4.8.6 at the start of the first article testing, and shall meet the capacity and number of cycles specified in the applicable specification sheet (see 4.4).

4.4.3 Definition of failure. The failure of a battery is defined as follows:

a. Failure to deliver the minimum electrical performance as specified in 3.11 and the applicable specification sheet.

b. Distortion, cracking or breakage which prevent the battery and any of its components from mating with a corresponding part or maintaining its electrical leak- proof requirements.

4.4.3.1 Degree of failure. The batteries shall be considered unacceptable if any of the sample battery numbers 1-6 fail to pass the inspection of Table III as specified in the applicable specification sheet.

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4.4.3.2 Rework and retest provisions. Once a failure has occurred, no rerun can be made of that test on the same or a new sample from the same group of samples unless positive action consisting of design change, modification, or rework actually related to observed failures has been applied to the group. Any subsequent rerun must start at the beginning of the test in which the failure occurred provided aforementioned action will not affect any tests prior to the test in which the failure occurred.

TABLE III. First article testing.

Req Para	Inspection	Battery No.	Test
3.4	Visual and Mechanical	1 to 6 1/	4.5.2.2
3.6	Dimensions and Weight	1 to 6 1/	4.6.6
3.10.2	Battery Voltage	1 to 6	4.5.2.3
3.11	Full Capacity Discharge (initial)	1 to 6	4.8.2
3.11	Cycle Life	5, 6	4.8.6
3.11	Overcharge	2,3	4.8.1.1
3.10.1	Electrical Leakage	1 to 6	4.10
3.11	High Rate	1, 3	4.8.3
3.11	Low Temperature (4.6.2)	2, 4	4.8.4
3.11	Retention of Charge	1,2,3,4	4.8.5
3.11	High Rate Pulse	1, 3	4.8.7
3.12	Thermal Shock (charge after this test)	1,2,3,4	4.9.1
3.12	Mechanical Shock	1,2,3,4	4.9.2
3.12	Vibration (discharge)	2, 3	4.9.3
3.13	Immersion	1,2,3,4	4.9.4
3.10.1	Electrical Leakage	1,2,3,4	4.10
3.11	Full Capacity Discharge	1,2,3,4	4.8.2
3.4.2, 3.15	Battery Case-Vent	1,4	4.11

1/ If final packaging is not available at time of the start of cycle life testing, samples 5 and 6 may be cycled in a simulated case and are not required to meet requirements 3.4 and 3.6.

4.5 Quality conformance inspection.

4.5.1 Quality conformance inspection. Inspection of product for delivery shall consist of groups A, B and C inspections. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion may perform all or any part of the specified inspection, to verify the

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contractor's compliance with specified requirements (see 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.2 Group A inspection. Each unit of each inspection lot shall be subjected to and meet the requirements of Tables IV and V utilizing inspection Level II of MIL-STD-105. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein or in the contract or purchase order shall be in accordance with MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

4.5.2.1 Disposition of group A samples. Unless otherwise specified, group A sample batteries shall be shipped and included in the quantity on order if they meet the necessary requirements.

Table IV. Group A inspection.

Inspection	Req para.	Test para.	AQL (%) Major	Minor
Visual and Mechanical Examination	3.4	4.5.2.2	1%	4%
Battery Voltage 1/	3.10.2	4.5.2.3	1%	-

1/ This defect is considered major.

4.5.2.2 Visual and mechanical. Samples shall be examined to verify that the basic materials, component materials and parts, design and construction, marking and workmanship are in accordance with all the requirements of 3.4.

4.5.2.3 Battery voltage. The battery at 23° ±°5 C (73.5°±9°F) shall be given a charge pulse at the 2C rate for 10 ±3 seconds. The voltage shall be constantly monitored and shall meet the requirements of 3.10.2.

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TABLE V. Classification of defects for batteries.

Major Category	Defects	Method of Insp
100	Zero (0) voltage.	Voltmeter
101	Contact surfaces obstructed by insulation compounds so that electrical use is affected.	Visual
102	Pitting blow holes or other imperfections in cell containers causing structural weakness.	Visual
103	Improper assembly causing parts to be inoperative or unsafe in service.	Visual
104	Weight not within specified limits.	Scale
105	Location and polarity of receptacle not as specified.	Visual
106	Terminal markings, identification label and operating instructions not as specified.	Visual
107	Corrosion which could cause mechanical, operational or electrical failure.	Visual
108	Particles of solder, flux or other foreign material which could cause mechanical, operational or electrical failure.	Visual
109	Insulators or insulation missing or damaged so as to cause electrical failure.	Visual
110	Free movement of connector restricted.	Visual
Minor		
201	Deformation or damage which does not affect operation or use in service.	Visual
202	Burrs or imperfections which do not interfere with proper use in operation, assembly or disassembly or cause an unsafe condition in service.	Visual
203	Contact surfaces of the terminals obstructed by insulation compounds, but would not cause mechanical or electrical failure in service.	Visual
204	Insulators or insulation damaged, but would not cause mechanical or electrical failure in service.	Visual

4.5.3 Group B inspection. This inspection, including sampling, shall conform to Table VI and to the special procedures for small sample inspection of MIL-STD-105. The AQL and inspection levels shall be as indicated in Table VI. Group B inspection shall normally be performed on lots that have passed Group A inspection and on samples selected from units that have been subjected to and

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met the Group A inspection. Samples for subgroup II may be selected from units that have passed subgroup I inspection.

4.5.3.1 Order of inspection within group B. Group B inspection shall be performed in that order shown in Table VI.

TABLE VI. Group B inspection.

Subgroup I	AQL	Insp Level	Req Para	Test Para
Dimensions and Weight	2.5%	S-2	3.6	4.6.6
Capacity Discharge	2.5%	S-2	3.11	4.8.2
High Rate Discharge	2.5%	S-2	3.11	4.8.3
Subgroup II				
Low Temperature Discharge	2.5%	S-1	3.11	4.8.4
Pulse Discharge	2.5%	S-1	3.11	4.8.7
Overcharge	2.5%	S-1	3.12	4.8.1.1
Electrical Leakage	2.5%	S-1	3.10.1	4.10

4.5.3.2 Disposition of conforming group B units. Unless otherwise specified, sample units which have been subjected to and passed Group B inspection may be shipped against the quantity on order provided any damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.4 Group C inspection. This inspection shall consist of the tests listed in Table VII. Shipment of batteries shall not be made until subgroup I and II have met the required inspections and 28 cycles of the required number of cycles of subgroup III have been successfully completed. Inspection shall be performed in the order shown.

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TABLE VII. Group C inspection.

Inspection Paragraph	Requirement Paragraph	Test Paragraph	Subgroups I	II	III
Thermal Shock	3.12	4.9.1	X	X	
Shock, mechanical	3.12	4.9.2	X		
Vibration	3.12	4.9.3		X	
Immersion	3.13	4.9.4	X	X	
Full capacity discharge	3.11	4.8.2	X	X	X
Retention of charge	3.11	4.8.5	X		
Cycle life	3.11	4.8.6			X
Electrical leakage	3.10.1	4.10	X	X	X
Battery Vent-Case	3.4.2, 3.15	4.11	X	X	

4.5.4.1 Sampling for group C inspection. Samples shall be selected for each subgroup I through III in accordance with Table VIII. The samples shall be selected at the start of the contract with the exception of contract lot sizes less than 500 batteries (see 4.5.4.1.1). These samples shall constitute the group C requirement for the first 10 percent of batteries produced.

4.5.4.1.1 Small contract lot size (see 6.5.1.). For those contracts or purchase orders whose total quantity on order is 500 or less, group C samples shall be selected at random from the entire production on the contract or purchase order with the exception of the cycle life test which may be selected from the first 15 units produced.

TABLE VIII. Group C sampling.

Contract lot size	Total no. of samples	Subgroup I	II	III	Intervals (approx % of production)
Up to 150	3	1	1	1	(See 4.5.4.1.1)
151 - 500	4	2	1	1	(See 4.5.4.1.1)
501 - 3200	6	1	1	1	10%
		1	1	1	40-60%
3201 -10000	8	2	1	1	10%
		1	1		20-40%
		1		1	50-80%
10001-35000	13	2	1	1	10%
		1	1	1	20-30%
		1	1	1	40-60%
		1	1	1	70-90%

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4.5.4.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract or purchase order.

4.5.4.3 Disposition of group C sample units. Sample units which have been subjected to group C inspections shall not be delivered on the contract.

4.6 Inspection conditions and equipment.

4.6.1 Normal temperature. Unless otherwise specified herein, all measurements and tests shall be made at 23 $\pm$ 5°C (73.4 $\pm$ 9°F) at ambient atmospheric pressure and relative humidity with a minimum storage time of 2 hours between charge and discharge except for cycle life test.

4.6.2 Low temperature. Low temperature charged storage and discharge tests shall be conducted at the temperature indicated in the applicable specification sheets, with a minimum storage time of 16 hours between charge and discharge. Tolerances of specified temperatures shall be kept with $\pm$ 1.1°C (2°F).

4.6.3 High temperature. High temperature charged storage and discharge tests shall be conducted at the temperature and for the length of time indicated in the applicable specification sheet. A minimum storage time of 4 hours at the specified temperature shall be required to achieve temperature stability before any high temperature charge or discharge or for a discharge following high temperature storage. Tolerances of specified temperatures shall be kept within $\pm$ 1.1°C (2°F).

4.6.4 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The contractor shall establish and maintain a system for the calibration of all measuring and test equipment in accordance with MIL-STD-45662.

4.6.5 Instrument accuracy.

4.6.5.1 Electrical indicating instruments. All voltmeters and ammeters shall be accurate within $\pm$ 1 percent of the full scale reading. The range shall be such that the readings are taken on the upper half of the scale. The sensitivity of voltmeters shall be at least 1000 ohms per volt.

4.6.5.2 Resistor and current tolerances. In all tests involving discharge through a resistor, such resistance shall be accurate within $\pm$ 0.5 percent. During the charging and discharging of batteries, conducted at various rates specified herein, the current shall be maintained within a tolerance of $\pm$ 3% of the specified value.

4.6.5.3 Timing. The timing of discharges one minute or less shall be maintained within 5 percent. All others shall be accurate within 1 percent.

4.6.6 Dimensions and weight. The battery shall be weighed and measured for conformance to the requirements of the specification sheets. A gauge or other suitable measuring device shall be

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used to determine dimensional conformance.

4.7 Elastomer material. All elastomer materials shall meet the applicable requirements of ASTM D 2000 and 3.8.2.

4.8 Electrical performance tests.

4.8.1 Charging. All batteries shall be charged by the constant current method at normal temperature conditions per 4.6.1 at the rates and for the length of time specified in the applicable paragraphs and specification sheet (see 3.1) except as provided for in paragraph 4.8.1.2.

4.8.1.1 Overcharge. Batteries shall be charged per 4.6.1 at the rate specified in the applicable specification sheet for the specified number of hours. Voltage readings shall be recorded and meet the requirements of 3.12.b. The battery shall then be discharged and meet the overcharge test of the applicable specification sheets. The battery shall be examined to insure compliance with all requirements of paragraph 3.12.

4.8.1.2 Rapid charging. When permitted in specific tests and approved by the Government, batteries may be charged by a quick charge method (2 hours or less) employing control methods relying on sensing changes in voltage with time (negative delta V) or equivalent.

4.8.2 Full capacity discharge. After being charged in accordance with 4.8.1 the battery shall be stabilized per 4.6.1. It shall then be discharged at the rate specified on the applicable specification sheet to the specified final voltage, and meet full capacity discharge test of the applicable specification sheet.

4.8.2.1 Initial full capacity discharge. For initial full capacity discharge, the battery, when subjected to the test specified in 4.8.2 shall meet the performance specified in the applicable specification sheet within any one of the first three cycles. Any cycle in which less than initial full capacity is achieved is not counted as part of cycle life requirement.

4.8.3 High rate discharge.

a. The battery shall be charged per 4.8.1 or per 4.8.1.2 and stored under the specified temperature and storage conditions.

b. The battery shall then be discharged at the rate specified in the applicable specification sheet.

c. The following shall be recorded:

(1) The terminal voltage, 5 seconds after the start of discharge.

(2) The elapsed time for the voltage to drop to the final voltage specified on the specification sheet after the start of discharge. The battery shall meet the high rate

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discharge test of the applicable specification sheet.

4.8.4 Low temperature discharge. Charge as specified in 4.8.1 or 4.8.1.2. The battery shall then be stored per 4.6.2. Discharge as specified at the low ambient temperature to the specified final voltage, and meet low temperature discharge test of the applicable specification sheet.

4.8.5 Retention of charge. The battery shall be charged as specified in 4.8.1. It shall then be stored under high temperature conditions (4.6.3) for the number of days specified in the applicable specification sheet and then returned to normal ambient temperature (4.6.1). It shall then be discharged at the rate specified on the applicable specification sheet to the specified final voltage and meet retention of charge test of the applicable specification sheet.

4.8.6 Cycle life test. The battery shall be tested as follows at the temperature specified in 4.6.1:

a. Cycle 1 - Charge per 4.8.1, discharge at $0.28C_5$ for 2.25 hrs.

b. Cycles 2-26 - Charge at C/5 for 3.75 hours or per 4.8.1.2, discharge at $0.28C_5$ for 2.25 hours.

c. Cycle 27 - Charge per b above. Discharge per 4.8.2.

d. Cycle 28 - Charge per 4.8.1, discharge per 4.8.2.

e. Check for electrical leakage (4.10).

f. Repeat cycles 1-28 until any of the following conditions occur:

(1) The capacity on any 28th cycle is lower than that specified on the cycle life test of the applicable specification sheet (see 3.1).

(2) The battery fails to pass the electrical leakage inspection of 4.10.

(3) The number of satisfactory cycles completed equals that specified on the applicable specification sheet.

4.8.7 Pulse discharge.

a. The battery shall be charged as specified in 4.8.1 or 4.8.1.2 and stored under the specified temperature and storage conditions (see 4.6.1).

b. The battery shall then be discharged at the rate and for the times specified in pulse discharge test of the applicable specification sheet until the end voltages specified are reached.

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4.9 Environmental tests.

4.9.1 Thermal shock. With the battery in a discharged condition, it shall be subjected to temperature shock per Method 503.2, Procedure 1 of MIL-STD-810 at the temperature specified in 3.9.2, except the time at the temperature extremes shall be 2 hours and diurnal cycling not required. The container shall be examined for breaks, cracks, or other defects resulting from this storage. The requirements of 3.12 shall be met.

4.9.2 Mechanical shock tests (see 3.12).The batteries in a charged condition shall be rigidly attached to the test carriage by the mounting screws and shock tested at 40g for an 18 millisecond pulse duration in accordance with test Method 516.2, Procedure 1, of MIL-STD-810. Three shocks per axis, on each of the three orthogonal axis, shall be applied.

4.9.3 Vibration (see 3.12). With the battery fully charged, it shall be tested in accordance with Method 201 for low frequency conditions and Method 204, Test Condition C, Part 2 only for high frequency conditions of MIL-STD-202, with the following exceptions:

a. The battery may be vibrated in one plane through both low and high frequencies before being changed to the next plane on the vibration table.

b. The low frequency vibration may be varied either uniformly or logarithmically between the limits of 10 and 55 Hz and return to 10 Hz. The entire frequency range from 10 to 55 Hz and back to 10 Hz shall be traversed in approximately one (1) minute.

c. The length of time for low frequency vibration in one plane shall be 90-100 minutes (270-300 minutes for the complete test).

d. The battery shall be discharged at the rate specified on the applicable specification sheet in each plane during the low frequency vibration and the discharge continued to the specified final voltage. The capacity specified in the applicable specification sheet shall be met (see 3.1).

e. The voltage shall be recorded on a recording voltmeter during both high and low frequency vibration, and shall meet the requirements of 3.12.

4.9.4 Immersion. The battery shall be tested for leakage by either one of the test methods below and shall meet the requirements of 3.13 and the applicable specification sheet (see 3.1).

a. Method One - The battery with terminals protected against electrolytic action shall be immersed to a depth of 3 feet in fresh water for 2 hours. After 2 hours of immersion, the battery shall be removed and wiped dry on exterior surfaces. There shall be no evidence of leakage.

b. Method Two - Pressurize the battery with a minimum of 1.3 psig of gaseous N_2 for a

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minimum of 2 hours. Release the pressure and immediately transfer the battery into an immersion vessel. The battery shall be immersed in an upright position to a depth of 3 inches in fresh water. Carefully observe the battery (particularly the seals) for a total of 5 minutes for indication of leakage. Leakage shall be indicated by the emission of gas bubbles from the interior of the battery.

4.10 Electrical leakage. Electrical leakage shall be performed where indicated in Table III, VI, and VII. Voltage readings shall be taken from the case or ground to either terminal and between all electrically isolated terminals with a voltmeter having a 1 Megohm impedance. For example, on a battery with two parallel sections measure voltage from pin 1 to pins 2 to 5, etc. The battery shall meet the requirements of 3.10.1.

4.11 Battery case-vent test. Unless specified otherwise (see 3.1), a fully assembled battery shall be inserted into a fixture designed to contain possible bulging by the front, back and side faces of the battery case (see applicable specification sheet for face references). The fixture shall allow the battery to expand up to .010 inches over the specified maximum allowable dimensions. The fixture shall not cover the top and bottom of the battery. An air line shall be tapped into the battery and the internal pressure of the battery shall be increased 1.0 psia every 60 seconds until the battery case vent opens. The battery shall meet the requirement of 3.4.2. The fixture with the pressurized battery shall be lifted or positioned to allow the battery to slide out through the bottom of the fixture. If the pressurized battery cannot slide out of the fixture, weights shall be placed on top of the battery. The total force required (weights plus the weight of the battery) to allow the battery to slide out of the mount shall not exceed 12 pounds, and the battery shall meet the requirements of 3.15.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the Acquisition Activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Battery, rechargeable sealed, nickel- cadmium is used in electronic and communications equipment, and for other power requirements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. When first article inspection and rough handling tests of equipment are not required.

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c. If first article inspection (see 3.2), first article test plans and test report(s) are required (see 4.4.1).

d. If certification of parts, materials, or components are required (see 3.3).

e. Actions required relative to group C inspection failure (see 4.5.4.2).

6.3 Definitions.

6.3.1 Fully charged batteries. Batteries shall be considered fully charged when charged in accordance with 4.8.1.

6.3.2 Discharged batteries. Batteries shall be considered discharged when a specified discharge rate was conducted until a specified final voltage was reached. For shipping purposes only the batteries may retain up to 10% of its capacity.

6.3.2.1 Discharge rate. The rate of discharge is defined as:

$$I = \frac{1}{t} C_5$$

where I = current in amperes
t = hourly rate of discharge
C_5 = rated ampere hour capacity

For example, the expression for the five hour discharge current for a 25 ampere hour battery will be $I = \frac{1}{5}(25) = 5$ amperes

6.3.3 Final voltage. The final voltage of the battery is the specified final voltage reached at the end of discharge.

6.3.4 Cycle. A cycle is a combination of a charge and a discharge.

6.3.5 Measured performance. The data obtained in conducting a test in accordance with section 4 of this specification, such as capacity or cycle life, shall be known as the measured performance.

6.3.6 Specified performance. The numerical performance requirement specified in a specification sheet covering parameters such as voltage, capacity, cycle life, storage life, etc., shall be known as the specified performance.

6.4 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount

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of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 Definition of lot size.

6.5.1 Contract lot. The contract lot is the total of all batteries (exclusive of the number of batteries required as samples) of any one type, delivered in one or more inspection lots, under the terms of any one contract.

6.5.2 Inspection lot. The inspection lot is the quantity of batteries (exclusive of the number of batteries required as samples) of any one type of any one code, and produced at any one place of manufacture on any one contract.

6.6 Test report data. The following requirement shall apply to First Article and Quality Conformance Test Reports: Whenever there is a service or ampere hour requirement, the following data shall be included:

a. Initial Open Circuit Voltage (IOCV)

b. Initial Closed Circuit Voltage (ICCV)

c. Closed Circuit Voltage and Current shall be recorded at regular intervals. Interval shall be Service Requirement or shorter.

d. The time elapsed between ICCV and end voltage (EV).

e. Record how low a battery is discharged. Particularly whenever near zero volts.

Date and time of day that any portion of test is initiated shall be recorded in test report including initiation of soak at test temperature.

When a requirement limit is specified, a final quantitative figure shall be recorded.

6.7 International Standardization. Certain provisions of this specification are the subject of International Standardization Agreement QSTAG 315. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

6.8 . First Article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard

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production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.9 Subject term (key word) listing.

Battery
Nickel Cadmium
Rechargeable
Sealed

6.10 Change from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - ER
Air Force - 99

Review activities:

Army - CR, AV, GL
Air Force - 80

Preparing activity:

Army - ER

(Project 6140-A780)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-49436B(ER)	93 12 10

3. DOCUMENT TITLE
Batteries, Rechargeable, Nickel Cadmium, Sealed

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Army Research Laboratory	(908) 544-3148	995-3148
c. ADDRESS (Include Zip Code) Electronics and Power Sources Directorate ATTN: AMSRL-EP-RD Fort Monmouth, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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